11b · Self-selected ad exposure, on real data — the Criteo experiment (IV · CausalPy)

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Runs in the `legacy` environment (pymc<6): `make env-legacy`, kernel `cmp-legacy`. First run downloads the 311 MB Criteo file into `~/.cache/cmp` (one-time; a seeded subsample is cached after that).

The business decision. Notebook 11's story, at industrial scale and with nothing simulated: **13.98 million real users** from Criteo's advertising incrementality tests (Diemert et al. 2018). Users who *saw* an ad visit the advertiser at hugely elevated rates — but the ad system *chose* whom to show ads to (bids, browsing, auctions), so exposure is **self-selected** and the naive comparison credits the ad with intent that was already there. We want the true causal effect of exposure, to set the **budget cap per exposed user**.

Why this dataset is a gift for IV

Criteo's incrementality tests randomize **eligibility**: a random 15% of users are locked out of the campaign entirely. That randomized `treatment` flag is a textbook **encouragement instrument** — we randomize a *nudge* toward exposure (eligibility), not exposure itself — for the self-selected `exposure`:

- **Relevance** — eligibility moves exposure (3.6% of eligible users end up exposed) — *testable*, first-stage F below;
- **Exclusion** — being *silently eligible* does nothing to your visiting behaviour unless an ad actually reaches you — *untestable*, stress-tested in Step 6;
- **Exogeneity** — eligibility is randomized by the test infrastructure — *holds by design*;
- **Monotonicity, for free**: ineligible users are **never** exposed (one-sided noncompliance), so "defiers" are impossible *by construction*, not by assumption — and the **LATE** (local average treatment effect — the effect for exactly the users the instrument moves) specializes to the **effect on the exposed** (ATT), exactly the population a budget cap is about.

What replaces the simulator's planted truth

Two real-data anchors. **(a)** At n = 13.98M the design-based **Wald estimate** (intent-to-treat ÷ first stage) is essentially free of sampling noise — we compute it once on the full file and grade every subsample estimate against it. **(b)** The features let us *photograph the confounding itself*: f0–f11 are balanced across the randomized arms but sharply imbalanced across exposure — the self-selection nb11 could only simulate, measured.

```
FAST=False  subsample=200,000 rows  Bayesian fit on 100,000 rows
```

2 · Load the real data (what replaces "simulate a ground truth")

`cmp.data.load_criteo_uplift()` downloads the full file once, then caches a **seeded, balanced subsample** — `N_ARM` rows from each randomized arm. Balancing the 85/15 arms is legitimate *because* the sampling key is the randomized flag itself (both the ITT and the first stage are conditional on `treatment`, so selecting on it biases nothing) and roughly doubles precision per row. The **data card**:

fact	value
population	13,979,592 users, assembled from several incrementality tests
randomization	`treatment` = eligible to be targeted (85%) vs locked out (15%)
self-selection	`exposure` = actually saw an ad: **3.6% of eligible, 0.0% of ineligible**
outcomes	`visit` (4.7% base) — primary; `conversion` (0.29%) — secondary, rarer & noisier
features	f0–f11, anonymized dense floats (used for balance checks, not needed for IV)
full-data anchor	Wald LATE on visit **+28.7 pp**, on conversion **+3.2 pp** (computed below)

Provenance check (cheap hygiene). The card's base rates are the published ones: Criteo's dataset release reports 13,979,592 rows, an 85/15 eligibility split, and ≈4.7% visit / ≈0.29% conversion base rates (Diemert et al. 2018 — full citation in §7), matching what the loader serves. So we are analyzing the dataset the paper describes, not a truncated or corrupted download — reconcile any loader against its source paper before believing anything downstream. The full-file anchor below remains the primary yardstick; this row is provenance, not validation.

The one modelling accommodation, inherited from nb11 and disclosed the same way: outcome and exposure are 0/1, and CausalPy's joint-MvNormal IV treats both equations as **linear-probability** (Gaussian) — fine for the point estimate and the error correlation ρ, so those are what we read.

```
subsample: 200,000 rows (100,000 eligible / 100,000 locked out)
P(exposed | eligible) = 0.0365   P(exposed | locked out) = 0.0000
  -> one-sided noncompliance, first-stage F = 3,793
visit rate: 0.0382 locked out vs 0.0493 eligible (ITT +1.11 pp)
full-data anchor (n=13.98M): ITT +1.034 pp, first stage 0.0360,
  Wald LATE +28.7 pp on visit

          f0     f1     f6  treatment  exposure  visit  conversion
0  24.059999  10.06  -5.12        0.0       0.0    0.0         0.0
1  25.240000  10.06 -13.58        1.0       0.0    0.0         0.0
2  23.750000  10.06 -10.76        0.0       0.0    0.0         0.0
3  12.620000  10.06   0.29        1.0       0.0    0.0         0.0
4  25.190001  10.06  -2.41        1.0       0.0    0.0         0.0
```

3 · Identify — a real instrument, and the confounding made visible

Exposure is endogenous: the ad system bids more for users who browse more, and browsing predicts visiting with or without ads. The instrument's three conditions are laid out in the header; before photographing the confounding, here is the estimand in symbols — because "the LATE specializes to the ATT" is this notebook's central identification claim, and it deserves a derivation, not a bullet.

The estimand: ITT, first stage, Wald — and why Wald = LATE = ATT here

Notation as in notebook 11 — Z the instrument, X the endogenous exposure, Y the outcome. (The print-outs' "Z→D" arrow is the same first stage: D, for "dose", is the econometrics alias for our X.) With $Z_i \in \{0,1\}$ the randomized eligibility, $X_i \in \{0,1\}$ actual exposure, $Y_i \in \{0,1\}$ a visit, potential outcomes $Y_i(x)$ and potential exposures $X_i(z)$, the three observed quantities are

$$\text{ITT} \;=\; \mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0], \qquad \pi \;=\; \mathbb{E}[X \mid Z{=}1] - \mathbb{E}[X \mid Z{=}0], \qquad \hat{\beta}_{\text{Wald}} \;=\; \frac{\text{ITT}}{\pi}.$$

Why the ratio is causal: write $Y_i = Y_i(0) + X_i\,(Y_i(1) - Y_i(0))$ and take expectations given Z. Randomization (exogeneity) plus exclusion make $\mathbb{E}[Y(0) \mid Z{=}1] = \mathbb{E}[Y(0) \mid Z{=}0]$, so the $Y(0)$ terms cancel from the ITT and only instrument-driven exposure survives:

$$\text{ITT} \;=\; \mathbb{E}\big[(Y(1) - Y(0))\,X \mid Z{=}1\big] \;-\; \mathbb{E}\big[(Y(1) - Y(0))\,X \mid Z{=}0\big].$$

In this design the second term is exactly **zero**: locked-out users are never exposed — $P(X{=}1 \mid Z{=}0) = 0$, printed as 0.0000 in step 2, a fact of the serving infrastructure, not a sample accident. So, with $\pi = P(X{=}1 \mid Z{=}1)$,

$$\hat{\beta}_{\text{Wald}} \;=\; \frac{\mathbb{E}\big[(Y(1) - Y(0))\,X \mid Z{=}1\big]}{P(X{=}1 \mid Z{=}1)} \;=\; \mathbb{E}\big[\,Y(1) - Y(0) \mid X{=}1\,\big] \;=\; \text{the ATT of the exposed}$$

(Bloom 1984) — the conditioning on $Z{=}1$ drops because *every* exposed user is an eligible user. In the general two-sided case this ratio is Imbens–Angrist's complier LATE and needs monotonicity (no defiers) as an *assumption*; one-sided noncompliance delivers it by construction, and the "compliers" are literally the exposed. A **defier** would be a user who gets exposed *only because* they were locked out — logically impossible here, since locked-out users are never exposed at all, which is exactly why monotonicity holds by construction. On the full file the pieces are ITT $\approx +1.03$ pp and $\pi \approx 0.0360$, giving $\hat{\beta}_{\text{Wald}} \approx +28.7$ pp — the anchor every subsample estimate below is graded against.

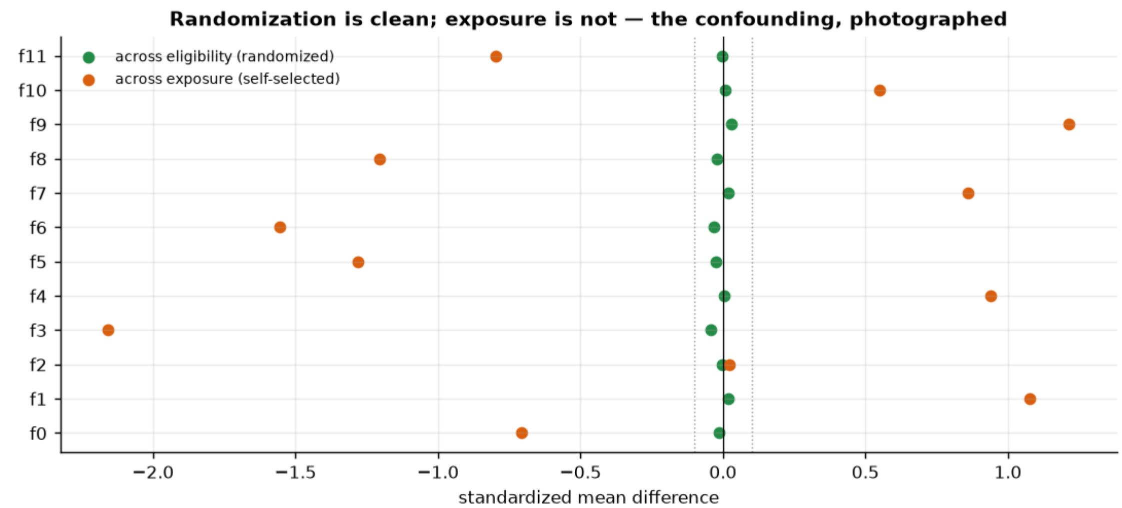
The confounding, photographed

And here is the part a simulator can never show. If randomization worked, the twelve user features must be **flat across the arms** (any imbalance would impeach the instrument's exogeneity). If exposure is truly self-selected, the same features must be **skewed across exposure**. Both facts are checkable — one chart, two stories:

- across **eligibility** (randomized): standardized mean differences $\approx 0 \to$ the instrument is clean;
- across **exposure** (chosen by the auction): large SMDs $\to$ the exposed *were different people before any ad effect* — the very confounder nb11 called `engagement`, photographed in real data.

This is also why no regression on f0–f11 could rescue the naive estimate: these are only the *observable* shadows of targeting; the auction also used signals not in the file (the unmeasured confounder), which is precisely the case where notebook 05's adjustment toolkit surrenders and IV takes over.

```
max |SMD| across eligibility: 0.042  (randomization holds)
max |SMD| across exposure:    2.157  (the exposed were different people to begin
with)
```



Step 0 · The classical read (no likelihood, no priors, no sampler)

Before any Bayesian machinery, ask what a competent analyst does here **without** it. Not a different analysis: the *same estimand* (§3's ATT of the exposed), identified by the *same* three assumptions (relevance, exclusion, exogeneity — monotonicity comes free from the one-sided lockout), estimated with the simplest tool that is still correct. The causal work lives in the identification argument, not in the machinery — and IV's machinery predates the computer, let alone the sampler (Wright, 1928).

Three regressions, and the reader must see all three naked, in this order:

$$\underbrace{Y = a_0 + a_1 X}_{\textbf{(1) as-treated / naive — BIASED}}, \qquad \underbrace{X = b_0 + \pi Z,}_{\textbf{(2) first stage}} \qquad \underbrace{Y = c_0 + \tau Z}_{\textbf{(3) reduced form = ITT}}.$$

Regression (1) splits users by the exposure they *received* — the attribution dashboard's number, and the one §3's SMD chart just impeached: the exposed were different people before any ad ran. Regressions (2) and (3) both regress on the **randomized** eligibility Z, so each is a clean experiment.

Neither is the answer on its own. The ITT $\hat{\tau}$ is unbiased for *the intent* — the effect of *being eligible* — but it is **diluted by non-compliance**: only ~3.6% of eligible users are ever actually shown an ad, so the lottery's effect on visits is averaged over the ~96.4% it never reached. The first stage $\hat{\pi}$ measures exactly that dilution. Divide, and the dilution cancels:

$$\hat{\beta}_{\text{Wald}} = \frac{\hat{\tau}}{\hat{\pi}} = \frac{\bar{Y}_{Z=1} - \bar{Y}_{Z=0}}{\bar{X}_{Z=1} - \bar{X}_{Z=0}} = \frac{\text{ITT — what eligibility did to visits}}{\text{compliance — what it did to exposure}} = \text{LATE} \overset{\text{one-sided}}{=} \text{ATT of the exposed}$$

The last equality is §3's Bloom (1984) result, derived there in potential outcomes and *computed* here for the first time. Units make the intuition: the numerator is in visits-per-user, the denominator in exposures-per-user; the ratio is **visits per exposure**, which is what a bid is priced in.

With one binary instrument, one binary endogenous regressor and no controls, two-stage least squares is exactly this ratio — `cl.iv_2sls` and $\hat{\tau}/\hat{\pi}$ agree to floating point, and the cell below *asserts* it rather than asking you to take it on faith (as notebook 11 does on its simulated twin). So why run 2SLS at all? For the two things a bare ratio cannot carry: a **standard error** and the **first-stage F**.

The interval: the delta method, because this is a ratio. Both $\hat{\tau}$ and $\hat{\pi}$ are noisy, so the noise of their ratio must be propagated. A first-order Taylor expansion of $\hat{\tau}/\hat{\pi}$ around the truth gives the **delta-method** variance:

$$\operatorname{Var}(\hat{\beta}) \approx \frac{1}{\hat{\pi}^2}\operatorname{Var}(\hat{\tau}) + \frac{\hat{\beta}^2}{\hat{\pi}^2}\operatorname{Var}(\hat{\pi}) - \frac{2\hat{\beta}}{\hat{\pi}^2}\operatorname{Cov}(\hat{\tau}, \hat{\pi}),$$

and $\hat{\beta} \pm 1.645\,\text{SE}$ is **the classical interval of this notebook** — the one §5's fold chart recomputes per fold and §5x holds the posterior against. The three terms are the numerator's noise, the denominator's noise, and their correlation (positive: a fold that happens to draw high-intent users gets *both* more exposure and more visits). Because $\hat{\pi}$ is pinned down so precisely here, the last two terms are nearly negligible: the code prints the one-term shortcut $\text{SE} \approx \text{SE}(\hat{\tau})/\hat{\pi}$ beside the full three-term SE so the approximation is **audited, not asserted** — and beside `cl.iv_2sls`'s own 2SLS standard error, which on this saturated binary design should land in the same place.

The covariance choice, and its one-line reason. A cross-section of users — no panel, no time series, so no clustering and no HAC. But every variable here is **0/1**: a Bernoulli's variance is $p(1-p)$, tied to its mean, so the errors are heteroskedastic *by construction* in all three regressions. All three therefore take **HC1** (heteroskedasticity-robust) standard errors, and the delta-method SE above is built from arm-specific variances, which is the same correction done by hand.

Relevance, measured — and this is the opposite of a weak-instrument problem. For a single instrument the first-stage F is the F-statistic of regression (2), $F = \frac{R^2/1}{(1-R^2)/(n-2)}$: the instrument's signal in the first stage relative to its noise. The folklore threshold is $F > 10$, and it has teeth — with one instrument, the 2SLS estimator's residual bias *relative to the OLS bias it is meant to remove* is roughly $1/F$, so $F = 10$ means "at most about a tenth of the OLS bias leaks back in". Here F is in the **thousands** (printed in §2 and again below): eligibility is randomized and mechanically enforced, so it is about as strong an instrument as a real dataset ever hands you. The cell cashes $1/F$ out in percentage points — the leakage is a rounding error. Say it plainly: *this notebook has no weak- instrument problem, and the diagnostic is here to prove that, not to* *worry about it.* (nb11's §5c shows what the same estimator does when F falls through 10; it is not a graceful degradation.)

And the referee is not a planted truth. On simulated data (nb11) Step 0 could be graded against the €15 we planted. Here nothing was planted. What replaces it is the **full-file anchor**: the same three quantities — ITT, first stage, Wald — computed *exactly*, design-based, on all **13,979,592 rows** (`cmp.data.criteo_full_anchor()`), where sampling noise is negligible at the precision we care about. It is not a truth, it is a *referee*: the number this same identification strategy converges to when you stop subsampling. Every classical estimate below is graded against it, and so is the posterior in §5x. That substitution — a full-population design-based estimate standing in for ground truth — is the honest real-data move, and it is worth more than most applied projects ever get.

```
(1) as-treated / naive  Y~X  [BIASED]: 0.3895  [90% CI 0.376, 0.4029]  (SE
0.00819, HC1 heteroskedasticity-robust, n=200,000)
(2) first stage        X~Z  (pi): 0.03654  [90% CI 0.03556, 0.03752]  (SE
0.000593, HC1 heteroskedasticity-robust, n=200,000)
(3) reduced form = ITT Y~Z  (tau): 0.01112  [90% CI 0.009616, 0.01262]  (SE
0.000915, HC1 heteroskedasticity-robust, n=200,000)


DILUTION, undone.  The ITT is unbiased for the INTENT but only 3.65% of eligible
  users were ever shown an ad, so it is diluted by that compliance rate:
  ITT +1.112 pp / pi 0.0365 = WALD +30.43 pp per EXPOSURE  (vs 2SLS +30.43 pp)
  -> identical to floating point: 2SLS *is* the Wald ratio here. The machinery
buys the SE and the F.
  Under one-sided noncompliance (§3, Bloom 1984) this is the ATT of the ~3.6% of
users the auction actually reaches.

THE CLASSICAL INTERVAL - Wald +30.43 pp, 90% CI [+26.44, +34.42] pp  (delta
method)
  three formulas for the SAME point estimate's SE, in pp:
    delta, three terms (numerator + denominator + covariance) : 2.425   <-
reported
    delta, shortcut    (numerator noise only; used per-fold in §5): 2.503
(+3.2%, conservative)
    cl.iv_2sls         (2SLS algebra)                           : 2.425
  -> the last two agree to x1.0000. On a saturated binary design the 2SLS
     variance and the hand-propagated ratio variance are the same object,
computed two ways.

RELEVANCE - first-stage F = 3,793   (cl.iv_2sls weak flag: strong - F is 379x
the F>10 bar)
  This is the OPPOSITE of a weak-instrument problem: eligibility is randomized
and mechanically
  enforced. 1/F cashed out: the naive number is off by +10.2 pp;
  the rule says ~1/F of that bias leaks back into 2SLS -> 0.0027 pp. A rounding
error.
```

```
GRADE vs the 13.98M-row anchor (Wald +28.70 pp; full-file naive +37.92 pp) - n =
200,000 here:
  (1) naive   +38.95 pp   off by +10.25 pp   anchor is OUTSIDE its 90% CI
[+37.6, +40.3] pp
      - the TIGHTEST interval here (SE 0.82 pp) around a number that is +10.2 pp
wrong.
        Precision is not accuracy: 13.98M more rows would only shrink it around
the same wrong number.
  (3) ITT      +1.11 pp   unbiased for the INTENT (anchor ITT +1.03 pp) - but it
is NOT the per-exposure price: it is diluted by the compliance rate, a factor of
1/pi = 27x.
  (4) Wald    +30.43 pp   off by +1.73 pp   anchor is INSIDE its 90% CI [+26.4,
+34.4] pp
      - the self-selection wedge the instrument removed: 8.5 pp per exposure,
        i.e. ~36% of the dashboard's number was intent we already owned.

What this 90% confidence interval does NOT say: that there is a 90% probability
the true effect lies inside it. It is a property of the *procedure* - 90% of
intervals built this way would cover the truth across repeated samples. This
interval either contains the truth or it does not. To get a probability *about
the effect itself* - the thing a go/no-go rule like P(lift > cost) >= 0.9
actually needs - you need a posterior. That is what the Bayesian section adds.
```

Read-out — the classical answer, in business terms. The rungs printed above are the whole of applied IV, and each one is a different *number a business could act on*:

1. **The naive as-treated gap is the dashboard's number, and it is wrong by a third.** It is also the **tightest interval in this notebook** — a 0/1 outcome on a large sample buys a standard error of a fraction of a percentage point — and it still misses the anchor by miles. That is the single most useful thing in Step 0: **precision is not accuracy.** No amount of extra data rescues it; the other 13.8 million rows only shrink that interval around the same wrong number (the full-file naive gap, printed above, is off by the same third). §3's SMD chart already told us why: the exposed *were different people*, and f0–f11 are only the observable shadow of a targeting rule we cannot see.
2. **The ITT is honest and useless on its own.** It is unbiased for the intent — randomization guarantees it — and it lands within a hair of the full-file ITT. But it is the effect of *eligibility*, averaged over the ~96% of eligible users the auction never actually reached, so as a **price per exposure** it understates by the reciprocal of the compliance rate, $1/\hat{\pi}$ — nearly thirty-fold (printed above). Reporting the ITT as "the ad's effect" is the mirror-image error to reporting the naive gap: one is diluted, the other inflated. Only their ratio is denominated in the unit a bid is priced in.
3. **The Wald ratio undoes exactly that dilution — and it is 2SLS.** The assertion in the code is the point: with a binary instrument and a binary exposure there is no daylight between "divide two experimental contrasts" and "run two-stage least squares". The estimator is a division. What 2SLS adds is bookkeeping: a standard error and an F.
4. **The F is a certificate, not a worry.** In the thousands — the instrument is randomized and enforced by the ad server, not scraped from behaviour. The 1/F cash-out prices the residual OLS-bias leakage at a rounding error. This is the regime IV was designed for and rarely finds.

And the grade is real. Every number was scored against the 13.98M-row anchor, not against a truth we planted — the printed `INSIDE/OUTSIDE` verdicts are the closest thing real data offers to a marked exam. (At the notebook's full working size the Wald's interval covers the anchor comfortably; on a small FAST slice the ratio's SE is several pp — the fold chart in §5 prices exactly that noise.)

Notice what has *not* happened yet. No likelihood, no prior, no sampler. The identification — a randomized, excluded, one-sided instrument — did all the causal work; three least-squares slopes and a division did all the estimation, in milliseconds. **This is the honest baseline the Bayesian section must justify itself against**, and §5x holds it to that with the anchor as referee.

The guardrail — what that confidence interval does *not* say (printed above by `cannot_say()`, in the same words in every notebook of this cookbook): it is **not** a 90% probability that the true exposure→visit effect lies inside it. A confidence interval is a property of the *procedure* — 90% of intervals built this way would cover across repeated samples — and this one either covers the anchor or it does not (the print says which). Here is exactly where that bites: §6 must answer *"what is the probability an exposure is worth more than the €c we would have to bid for it?"* — $P(v\beta > c) \geq 0.9$. That asks for a probability **about the effect itself**. The classical apparatus cannot supply one, and — the part people get wrong — **bootstrapping the Wald ratio would not fix it**: the share of resamples above c is a frequency over hypothetical *datasets*, not a belief about β. That quantity does not exist in the classical vocabulary. It exists in a posterior. That, and not a better point estimate, is what §4 is for.

4 · Estimate — the Bayesian IV model

Step 0 already produced a causal number: the Wald ratio, which *is* 2SLS, with a delta-method interval and a first-stage F in the thousands to certify the instrument. What follows does **not** change the identification by a comma — same instrument, same assumptions, same ATT of the exposed. It changes the apparatus for expressing uncertainty about that number, and it estimates one object the classical route structurally cannot hand you: **the endogeneity itself**.

The Bayesian IV (CausalPy) fits the first stage and the outcome equation **jointly**, letting their errors be correlated — the same joint Gaussian as nb11's, with X = exposure and Y = visit:

$$\begin{pmatrix} X_i \\ Y_i \end{pmatrix} \sim \mathcal{N}_2\left(\begin{pmatrix} b_0 + \pi Z_i \\ \beta_0 + \beta X_i \end{pmatrix},\ \Sigma \right), \qquad \Sigma = \begin{pmatrix} \sigma_X^2 & \rho\,\sigma_X \sigma_Y \\ \rho\,\sigma_X \sigma_Y & \sigma_Y^2 \end{pmatrix},$$

where β is the causal effect the budget cap needs — the same β Step 0 just estimated by division — and the off-diagonal ρ **is** the endogeneity: the correlation between the exposure- and visit-equation errors that hidden intent induces and that the naive comparison ignores. 2SLS *removes* the endogeneity by projection; it never names it and has no ρ to report. The joint model *estimates* it. Two reading notes: ρ is small here even at full fit size — visits are rare events, so even strong targeting induces only a modest error *correlation* — and a small subsample may not resolve its sign at all. The naive-vs-IV gap, not ρ's magnitude, is where the bias shows.

Priors. $b_0, \pi, \beta_0, \beta \sim \mathcal{N}(0, 50^2)$ — that is the `sigmas: [50, 50]` passed in the code, the weakly-informative override of CausalPy 0.8.1's default priors, which centre on the 2SLS point with $\sigma = 1$ and yield a band roughly 3× too narrow (nb11 demonstrates that band excluding its planted truth; same model, same library version here, so the gotcha transfers verbatim) — plus a mild **LKJ** prior on the error correlation, $\Sigma \sim \text{LKJCholeskyCov}(\eta = 2)$ ($\eta = 2$ gently pulls ρ toward 0, so the data has to argue for endogeneity). With ~100,000 rows the likelihood swamps any of this; §5x measures exactly how little the prior is doing.

The fit size, and the linear-probability accommodation. The fit uses `N_FIT` rows — NUTS (the Hamiltonian sampler underneath) costs scale with n, and Step 5 grades this subsample posterior against the 13.98M-row anchor to show what the smaller n costs. Outcome and exposure are 0/1 while the model is Gaussian, so both equations are **linear-probability**. One upgrade on that disclosure: because Z and X are both binary, both **conditional means** above are *saturated* — a linear function of a binary regressor can match $\mathbb{E}[X \mid Z]$ and $\mathbb{E}[Y \mid X]$ exactly — so the LPM approximation lives entirely in the Gaussian *likelihood*, not in the mean structure. That is precisely why the point estimate and ρ are the trustworthy read-outs, and the equation-level posterior widths are the thing not to over-read.

> **Reading the sampler's health check (the `IV convergence:` line below, and any PyMC warning beside it).** Three numbers say whether the MCMC sampler — the algorithm drawing from the posterior — converged: **R-hat** compares the variance *within* each chain to the variance *across* chains (1.00 is perfect, $\leq$ **1.01 is the usual pass bar**); **ESS** (effective sample size) is how many *independent* draws our autocorrelated chains are worth (a few hundred is ample for a posterior mean or interval); **divergences** are steps where the sampler broke down and silently distrusts that region (**you want 0**). Under this notebook's FAST profile the chains are deliberately short (200 draws over 2 chains), so R-hat can drift to ≈1.02 and PyMC may print an ESS / "problems during sampling" notice — a benign small-draw-count artifact that a FULL run (4 chains, more draws) clears. Here, at FULL (4 chains, 1,000 draws) the health check below prints a clean pass — R-hat at/near 1.01, ESS in the thousands, and 0 divergences. The point estimate and its interval are stable regardless, which the disjoint-fold check (Step 5) and the 13.98M-row anchor confirm.

```
Step 0, for reference:  Wald = 2SLS +30.43 pp  [90% CI +26.44, +34.42] pp   (n =
200,000)
Bayesian IV (100,000 rows) +29.37 pp   90% credible interval [+23.79, +34.95] pp
full-data anchor (13.98M)   Wald +28.70 pp
IV convergence: max r-hat 1.010 - min ESS 1201 - divergences 0
estimated endogeneity rho = +0.06 [90% +0.03, +0.10] - positive rho is the self-
selection: the same hidden leanings push a user into both exposure and visiting
```

5 · Validate — against 13.98 million rows

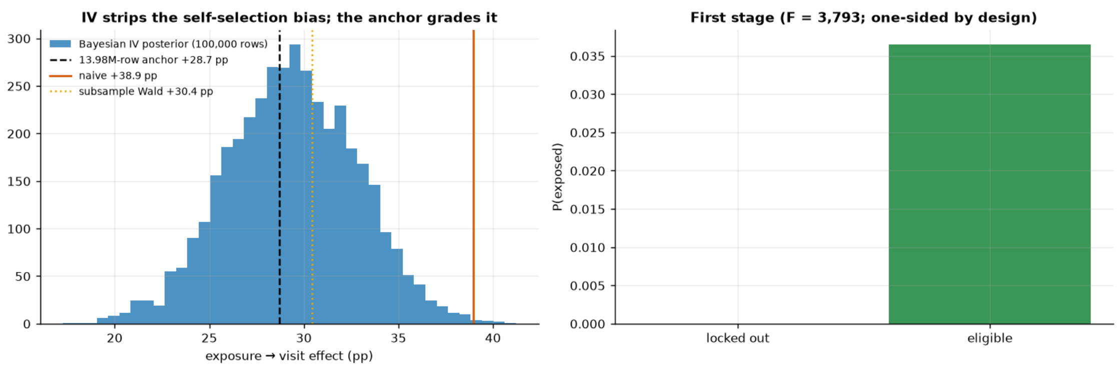
No planted truth, but a better yardstick than most real projects ever get: the full-file **Wald anchor** Step 0 already graded the classical rungs against, whose own sampling noise at n = 13.98M is negligible for our purposes. Three checks, plus the star exhibit:

1. **Anchor recovery.** The subsample Bayesian IV's 90% interval should cover the anchor (+28.7 pp), and its point should sit within a couple of pp — that is the real-data analogue of nb11's "recover the €15." (Expect the pass at the full working size; on a small FAST slice the Wald ratio's SE is several pp, so the interval can sit an SE or two off the anchor — subsampling noise that check 3 prices, not a failed identification.)
2. **Classical vs Bayesian on the same rows.** They are the same identification strategy estimated two ways, so they should agree; a gap would flag a modelling artefact, not a causal discovery. The head-to-head — Step 0's Wald and its delta-method interval against the

posterior, both scored on the anchor — is **§5x** below, at the end of this section.

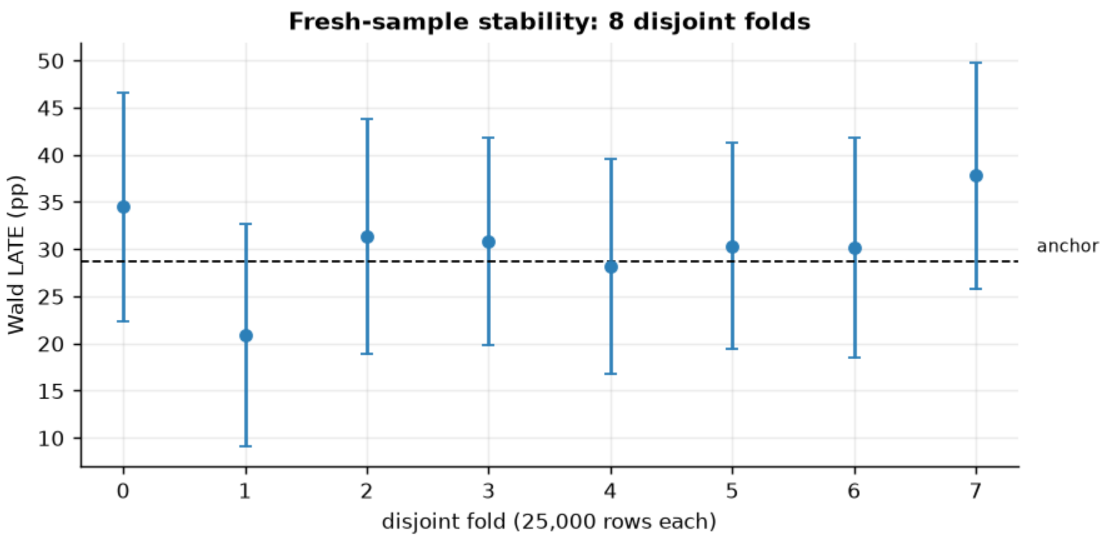
3. **Disjoint-subsample stability** — nb11's multi-seed loop, real-data edition: we can't refit on fresh *worlds*, but 13.98M rows let us refit on fresh *samples*. Split the working subsample into disjoint folds, compute each fold's Wald ± its **Step-0 delta-method interval** (the one-term shortcut, whose ~3% conservatism Step 0 audited against the full three-term SE — the licence to drop the denominator's noise is the enormous F: the relative noise of $\hat{\pi}$ is exactly $1/\sqrt{F}$, a fraction of a percent here, versus double-digit relative noise in a fold-level ITT). Check the spread, and how often those intervals cover the anchor.
4. **The naive − IV gap *is* the measured self-selection bias**: the euros an attribution dashboard would over-credit to the ad. On the full file the gap is deterministic (both full-file numbers are printed by Step 0 above): naive ≈ +37.9 pp vs the +28.7 pp anchor — the dashboard number runs ≈32% too big on the full file (the subsample re-measures it with its own slice's Wald noise, priced by the fold check).

```
anchor recovery: 90% interval [+23.8, +35.0] pp COVERS the 13.98M-row anchor
(+28.7 pp)
self-selection bias, measured against the anchor: naive +38.9 -> causal +28.7 pp
(+10.2 pp of pure targeting, ~36% inflation; this slice's own IV: +29.4 pp)
```



How to read this. *Left* — four vertical markers tell the story. The **orange** line is the naive exposed-vs-unexposed gap: the attribution-dashboard number, the one everything else should sit below. The **blue** histogram is the Bayesian IV posterior for the same causal question; the **gold dotted** line is the subsample Wald (same identification, frequentist arithmetic); the **dashed black** line is the 13.98M-row anchor (+28.7 pp). The horizontal distance from the orange line down to the anchor *is* the self-selection premium, measured: on the full file ≈9 pp of the ≈38 pp naive "effect" — roughly a third — is intent the exposed users already had. The printed checks grade the rest: the 90% interval covering the anchor is check 1, and posterior mean ≈ Wald is check 2 (§5x) (a gap there would flag a modelling artefact, not a causal discovery). (One number to not trip on: the naive gap *printed* here is this subsample's (+38.9 pp), a hair above the full-file +37.9 pp quoted in the text and printed in Step 0 — same story, different n.) One honesty note before over-reading the blue centre: with a 3.6% exposure rate in the denominator, subsample Wald ratios are genuinely noisy, so on a small slice the posterior can drift an SE or two from the anchor — that drift is sampling noise, priced explicitly by the fold chart below, not a broken instrument. *Right* — the first stage is the design itself, not a diagnostic that happened to pass: the locked-out bar is exactly zero (one-sided noncompliance — the fact that turned LATE into ATT in §3) and the eligible bar (≈3.6%) is the entire lever the instrument pulls. $F \gg 10$ is the licence for dividing by it; with a weak first stage the left panel would be meaningless.

```
fold Walds: mean +30.5 pp, sd 4.9 pp (anchor +28.7); 90% CIs cover the anchor in
100% of folds
the spread is the honest price of subsampling a rare exposure (3.6%) - and why
the anchor was computed on all 13.98M rows
each fold's interval is Step 0's delta-method CI (one-term shortcut), recomputed
on the fold
```



5x · Point estimate vs posterior — what the Bayesian layer actually bought

Step 0's Wald/2SLS and §4's posterior target the **same estimand** (the ATT of the exposed) under the **same assumptions**, and the table below puts them on the **same rows** — it re-runs the classical estimator on the exact N_FIT slice the sampler saw, so the comparison is apparatus versus apparatus and not n versus n. Both are then scored against the 13.98M-row anchor. Then we ask each one the question §6 actually needs answered.

Set expectations honestly before reading it. At this n — a hundred thousand rows, not three thousand — the likelihood swamps any weakly-informative prior, and the Bernstein–von Mises heuristic says the posterior should be *approximately the sampling distribution of the classical estimator*: same centre, same width. If Bayes buys precision or accuracy here, something is wrong. The interesting column is the last block, not the first.

```
SAME estimand (ATT of the exposed), SAME assumptions, SAME 100,000 rows.
Referee = the 13.98M-row anchor: +28.70 pp.
 (Rows 1-2 carry 90% CONFIDENCE intervals, row 3 a 90% CREDIBLE interval -
  printed in the same column, but NOT the same object.)

estimator (pp/exposure)         est     5%     95%   |err|   covers
naive as-treated [BIASED]     38.77  36.85  40.69   10.07       NO
```

```
Wald = 2SLS (Step 0, delta)  29.33  23.60  35.07   0.63     yes
Bayesian IV posterior (§4)   29.37  23.79  34.95   0.67     yes

LOCATION - the two causal estimators agree to 0.04 pp (0.01 classical SEs).
  Closer to the anchor: 2SLS (classical) - |err| 0.63 vs 0.67 pp,
  a distance of 0.01 SEs: noise, not skill.
WIDTH    - half-width +/-5.73 pp (delta method) vs +/-5.58 pp
  (posterior): a ratio of 0.97x. At n = 100,000 the likelihood swamps
  the N(0,50^2) prior, so the posterior IS (to within Monte-Carlo error) the
  sampling distribution the delta method wrote down in closed form. The sampler
  took ~20 minutes; the division took microseconds. Bayes bought no precision
  here - and at this n it never could.

P(exposure pays) = P(v*beta > c) at v = EUR 1.00/visit - §6's rule:
  c (EUR per exposure)         classical   Bayesian IV
  0.20                       not defined         1.00
  0.28                       not defined         0.66
  0.32                       not defined         0.23
  ('not defined' is not a dodge. A confidence interval carries no probability
about a
   hypothesis, and a bootstrap of the Wald ratio would not fix it: the share of
resamples
   above c is a frequency over hypothetical datasets, not a belief about beta.
Section 6's
   rule -- bid up to c while P(v*beta > c) >= 0.90 -- needs the right-hand
column, and the
   euro cap it reads off IS a posterior quantile: c* = v*q_0.10(beta).)

And the endogeneity itself: rho = +0.06 [90% +0.03, +0.10] - an interval on the
very
  self-selection that inflated the naive row. Step 0 has no rho to report, at
any width:
  2SLS projects the endogeneity away without ever naming it.
```

The honest verdict — what the posterior bought, and what it did not.

1 · They do not merely agree; they coincide. The table prints it: Step 0's Wald and §4's posterior mean land within a small fraction of one standard error of each other, and their intervals are the same width to within a few percent. This is not a happy accident, it is a theorem showing up in a print-out. At `N_FIT` $= 100{,}000$ rows the likelihood is overwhelming and the $\mathcal{N}(0, 50^2)$ priors are, relative to it, flat; Bernstein–von Mises then says the posterior *must* converge to a Gaussian centred on the efficient classical estimate with the classical sampling variance. The delta-method SE and the posterior standard deviation are estimating the same number, and they agree. **The sampler spent ~20 minutes rediscovering, by simulation, a division that took microseconds.**

2 · Which one is closer to the anchor is a coin toss — and the print says so. Both sit slightly above the 13.98M-row referee, both cover it, and the difference between their errors is a hundredth of a standard error. Do not read a winner into that gap: it is Monte-Carlo noise on the posterior plus one slice's sampling noise, not evidence that either apparatus is more accurate.

(Contrast nb11, where on 3,000 simulated rows the classical 2SLS was *genuinely* the closer of the two, because the Bayesian model's Gaussian first stage was committing a real linear-probability modelling error the two `lstsq` calls could not make. That error has not gone away here — it has been *drowned.* At Criteo's n, model error of that size is invisible next to nothing at all.) The one row that is decisively wrong is the same one as ever: the naive as-treated gap, carrying the **tightest** interval in the table around a number ~10 pp from the anchor.

3 · So say it bluntly: at this n, Bayes buys nothing on estimation. Not accuracy (the errors are indistinguishable). Not precision (the credible interval is, to within a rounding error, the confidence interval). Not the instrument diagnostic (the F is a classical object; the posterior would have gone on producing a beautifully smooth, entirely fictitious credible band as F fell through 10). Not robustness (the LPM approximation is the *Bayesian* arm's liability, not the classical one's). Anyone who tells you the Bayesian machinery *rescued* this analysis is selling something. On thirteen million rows of a randomized, mechanically-enforced instrument, the causal work was done entirely by the identification, and the arithmetic was a division. **This is a chapter where the classical arm wins on estimation, and the cookbook says so.**

4 · What it does buy — and it is not nothing: the decision. Look at the $P(v\beta > c)$ block. The classical column reads *not defined*, and that is the literal truth rather than a rhetorical concession: the frequentist apparatus attaches no probability to a hypothesis about a fixed parameter, and bootstrapping the Wald ratio would not fix it — the share of resamples above c is a frequency over hypothetical *datasets*, not a belief about β. Yet §6's rule is *nothing but* such a probability: **bid up to c while** $P(v\beta > c) \geq 0.9$, and the budget cap it reads off is literally a posterior quantile, $c^* = v\, q_{0.10}(\beta)$, denominated in euros. Every euro figure in §6 — the cap, the coin-flip price, the BUY/HOLD call, the exclusion stress test's flip threshold — is computed from that right-hand column and could not have been computed from the left. Second, ρ: the endogeneity itself, with an interval on it. 2SLS projects the self-selection away without ever naming it; on real data, where there is no answer key, a ρ that is credibly positive is *evidence* that the dashboard number was inflated — evidence the classical arm cannot produce.

The lesson, stated so it survives being quoted. At Criteo scale the Bayesian layer is a **convenience, not a rescue**: it does not improve the number, it changes what you are licensed to *say* about the number, and it propagates that licence coherently into euros. Where Bayes genuinely earns its compute is the opposite regime — thin data, hierarchical structure, real prior information, a model that must be built rather than divided (nb09's partial pooling, nb06's MMM). Here the honest reason to fit it is the decision rule in §6, and if you did not need $P(v\beta > c)$, Step 0 would have been a complete and defensible analysis on its own.

6 · Decide, in euros — the budget cap, and the exclusion stress test

The cap logic of nb11, with real numbers: an exposed user visits with ≈ **+29 pp** extra probability (the 13.98M-row anchor: +28.7 pp), so at an assumed **€1 value per incremental visit** an exposure is worth ≈ €0.29 — and the *naive* number (full-file exposed-vs-unexposed gap, ≈ +37.9 pp, printed in Step 0) would have priced it at ≈ €0.38. The decision rule: **BUY exposure while P(value × LATE > cost) ≥ 0.9**; sweeping the cost turns that into the **maximum viable price per exposed user**. Both business inputs are assumptions and get swept: the €1 visit-value scales the cap linearly, and the secondary `conversion` outcome gives an independent cap for a margin-per-conversion framing.

The exclusion stress test. The lockout is silent, but eligibility could still leak around the `exposure` flag — e.g. eligible users win *other* auctions differently, or "exposed" mislabels a below-the-fold impression. As in nb11: give eligibility a hypothetical direct effect δ on visits, correct the reduced form, and find the δ that would flip the BUY.

The rule, the cap, and the stress test, in symbols. With v the value per incremental visit, c the cost per exposed user, and $\beta^{(s)}$ the posterior draws of the exposure effect, the decision rule and the two prices the sweep reads off are

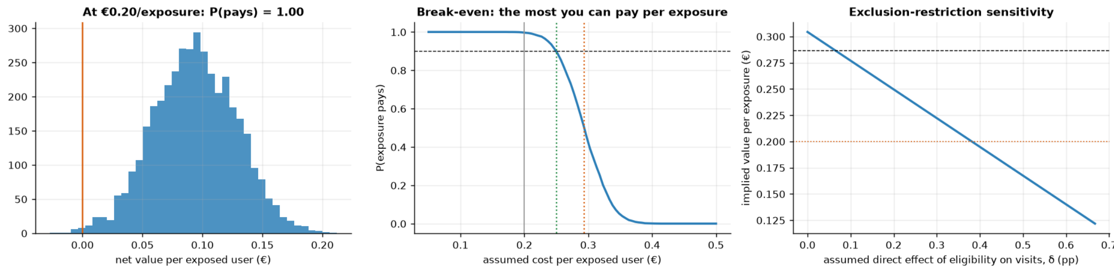
$$\text{BUY while } P(v\,\beta > c) \geq 0.9; \qquad c^* = v\,q_{0.10}(\beta), \qquad c_{50} = v\,q_{0.50}(\beta),$$

where $q_p(\beta)$ is the posterior p-quantile. Sweeping c against a fixed posterior just walks its quantile function — $P(v\beta > c) \geq 0.9 \iff c \leq v\,q_{0.10}(\beta)$ — so the budget cap c^* *is* a posterior quantile denominated in euros, c_{50} is the coin-flip price, and the cap a dashboard would set is $v\,\widehat{\beta}_{\text{naive}}$. For the exclusion stress: if eligibility leaks a direct effect δ (in visit probability) around exposure, the honest reduced form is $\text{ITT} - \delta$, so the corrected effect and the flip threshold are

$$\beta(\delta) = \frac{\text{ITT} - \delta}{\pi}; \qquad v\,\beta(\delta) > c \iff \delta < \delta^* = \text{ITT} - \frac{c\,\pi}{v}.$$

The flip threshold δ^* has a closed form — no grid search needed — and the code below prints it next to the grid scan of the third panel, so the two agree in front of you (up to the grid's resolution). Note $\beta(\delta)$ re-solves the ratio at point estimates: carry the posterior interval's width alongside it when briefing the cap, as in nb11.

```
at €0.20/exposed user and €1.00/visit: net €+0.09, P(pays) = 1.00
  -> BUY (to the exposed margin)
budget cap: clears the 90% bar up to €0.25/exposed user; a coin flip at €0.29
  - the naive number would have set the cap at €0.39 against the
  anchor's causal €0.29: 36% too high
conversion framing (secondary): full-data anchor LATE +3.20 pp
  -> at €40 margin/conversion an exposure is worth ~€1.28; the subsample
  pins it only to +3.17 pp (90% CI [+2.16, +4.19] pp
  -> €+0.86 to €+1.68) - at a 0.29% base rate this rung is honestly
  noisy; re-derive with the firm's own margin
exclusion stress: the BUY at €0.20 survives until eligibility's direct effect
  exceeds delta ~ 0.39 pp - 35% of the whole reduced form (+1.11 pp)
  would have to bypass exposure
  closed form agrees: delta* = ITT - c*pi/v = 0.38 pp (grid scan: 0.39 pp)
campaign scale: the naive cap embeds €0.10/exposure of pure targeting premium
  (naive €0.39 vs causal anchor €0.29); at 1,000,000 exposures/quarter
  that is ~ €102,458/quarter paid for intent the users already had
  full-file version (the CMO memo's numbers): naive cap €0.38 vs causal €0.29 ->
€0.09/exposure premium, ~ €92,164/quarter
```



Read-out — the three panels, then the sentence for the CMO. *Left*: the net-value posterior $v\beta - c$ at the assumed €0.20 cost; the printed P(pays) is its mass above the orange zero line, and the BUY/HOLD call is just whether that mass clears 0.9. *Middle*: the sweep is the posterior's quantile function in euros (§6's $c^* = v\, q_{0.10}(\beta)$) — where the curve crosses the dashed 0.9 line is the green budget cap, the orange dotted line is the coin-flip price $v\, q_{0.50}(\beta)$, and the grey line marks today's assumed cost. *Right*: exclusion stress — the implied value falls linearly as the assumed leak δ grows (§6's $\beta(\delta) = (\mathrm{ITT} - \delta)/\pi$); the printed δ^* is where the line crosses the orange cost line, and the dashed reference is the anchor's ≈€0.29.

What you would tell the CMO. "An exposure to the users this auction actually reaches is worth about €0.29 in expected incremental visits (13.98M-row anchor: +28.7 pp at our assumed €1/visit); cap bids at the printed 90%-confidence cap, a few cents below that. The attribution dashboard prices the same exposure at ≈ €0.38 (full-file naive gap: +37.9 pp) — about a third of that is intent we already owned, so a dashboard-calibrated cap overpays on *every* exposure: ≈ €0.09 each, order of €90k per quarter at a million exposures a quarter (the print above re-computes this at this run's numbers). Three scope lines before this goes into a rate card: it is the ATT of the ~3.6% of users the auction actually reaches — it prices *these* exposures and does not license doubling reach, because the next users the auction would find are, by its own logic, worse prospects; the €1 visit-value and €40 margin are swept assumptions, so finance should re-derive the cap with its own numbers; and the BUY survives an eligibility side-channel only up to the printed δ^* — a leak bigger than that flips the call."

7 · Caveats

- **ATT of the exposed, not everyone.** One-sided noncompliance makes monotonicity free, but the estimate speaks only for the ~3.6% of users the auction actually reached — the right population for "what is an exposure worth," the wrong one for "what if we doubled reach" (the next users the auction would reach are, by its own logic, worse prospects).
- **Exclusion is the load-bearing untestable.** The lockout is silent, but any eligibility side channel (auction dynamics on other campaigns, mislabelled below-the-fold "exposures") leaks into the estimate; step 6 priced the leak that would flip the call.
- **Linear-probability approximation** — binary outcome and binary exposure in a Gaussian joint model, as in nb11: read the point and ρ, not the equation-level posterior widths.
- **Visits ≠ profit.** The €1/visit and €40 margin/conversion are finance inputs, swept but assumed; the conversion-based rung is noisier (0.29% base rate) and its cap should be re-derived at decision time with the firm's own margin.
- **A composite of several tests** — Criteo assembled multiple incrementality tests; the LATE is a blend across campaigns/periods (fine for a benchmark cap, blur it into one campaign's

plan with care).
- **Subsampling is disclosed, not hidden**: every number above is grade-able against the full-file anchor, and the fold check shows exactly how much noise a 200k-row view carries.

Cite the data: Diemert, Betlei, Renaudin & Amini (2018), *A Large Scale Benchmark for Uplift Modeling*, AdKDD. Dataset: Criteo AI Lab, research use.